Ananya Omanwar

Curriculum Vitae

Department of Mathematics and Statistics
Queen's University, Kingston, Canada
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Education

2022–present
Ph.D. in Mathematics
Queen's University, Kingston, Ontario, Canada
- Supervisors: Fady Alajaji and Tamás Linder.
- Research area: generalized information measures, information-theoretic bounds on excess minimum risk, and Sibson capacity of channels with memory.

2019–2022
MSc in Mathematics
Indian Institute of Science Education and Research Mohali, Mohali, India
- Advisor: Neeraja Sahasrabudhe.
- Thesis: "Influencing Opinion Evolution over Finite Time Horizons."

2016–2019
BSc (Honours) in Mathematics
St. Stephen's College, University of Delhi, Delhi, India
- Graduated with First Division.

Research Interests

Generalized information measures; channel capacity and symmetry; channels with memory; information-theoretic bounds in statistical decision problems; feedback communication.

Publications and Manuscripts

[1] A. Omanwar, F. Alajaji, and T. Linder. *On the α-Sibson Capacity of α-Quasi-Symmetric Channels.* Revised manuscript under review at IEEE Communications Letters, 2026.

[2] A. Omanwar, F. Alajaji, and T. Linder. *Bounds on the Excess Minimum Risk via Generalized Information Divergence Measures.* Entropy, 27(7), article 727, 2025. doi:10.3390/e27070727.

[3] A. Omanwar, F. Alajaji, and T. Linder. *Bounding Excess Minimum Risk via Rényi's Divergence.* Proceedings of the 2024 International Symposium on Information Theory and Its Applications (ISITA), pp. 59–63, 2024. IEEE Xplore.

[4] S. Moharir, A. S. Omanwar, and N. Sahasrabudhe. *Diffusion of Binary Opinions in a Growing Population with Heterogeneous Behaviour and External Influence.* Networks and Heterogeneous Media, 18(3), pp. 1288–1312, 2023. doi:10.3934/nhm.2023056.

Work in Progress

- A. Omanwar, F. Alajaji, and T. Linder. *The α-Sibson Capacity of a Noise-Erasure Channel with Memory.* Working paper, 2026.

Talks and Presentations

November 2024
"Bounding Excess Minimum Risk via Rényi's Divergence"
2024 International Symposium on Information Theory and Its Applications (ISITA), Taipei, Taiwan

Awards and Funding

2022–present **Queen's Graduate Award**
Queen's University

2022–present **Queen's University Research Fellowship**
Queen's University

2019–2022 **Institute Fellowship (Integrated MS–PhD Programme)**
IISER Mohali

2019 **Vice-Chancellor's Gold Medal**
University of Delhi

Summer 2018 **Summer Research Fellowship**
Indian Academy of Sciences; hosted at IISER Pune

2017–2018 **Prof. Nagpaul Fellowship**
University of Delhi

Academic Service and Leadership

2023–2025 **President, Graduate Math Society**
Queen's University

- Represented graduate students at departmental meetings and organized social events.

Service as a Referee

- IEEE Transactions on Information Theory.
- International Symposium on Information Theory and Its Applications (ISITA), 2026.
- IEEE Information Theory Workshop (ITW), 2026.

Teaching Experience

University appointments at Queen's University; private tutoring undertaken independently.

Instructor

Winter 2026 **APSC 172: Engineering Calculus**
Teaching Fellow

- Taught sections 101–103; contributed to course coordination, assessment, office hours, and examination grading.

Teaching Apprenticeship

Winter 2024 **MATH 211: Algebraic Method**
Under Ernst Kani

Tutorial Instruction

Fall 2023, 2024 **MATH/MTHE 474/874: Information Theory**
Teaching Assistant

- Led weekly tutorials for approximately 75 students in 2023 and 60 students in 2024.

Fall 2026 **MATH/MTHE 351: Probability I**
Teaching Assistant

- Led weekly tutorials for approximately 75 students in 2026.

Marking and Grading

2022–2026 **Teaching Assistant**

Calculus, Linear Algebra, Differential Equations, Information Theory, and Probability

- Calculus: MATH 120, 121; APSC 171, 172.
- Linear Algebra: MATH 110, 111; APSC 174.
- Ordinary Differential Equations: MATH/MTHE 225; MATH 237.
- Information Theory; Data Compression and Source Coding: MATH/MTHE 474/874.
- Probability: MATH/MTHE 351.
- Marked assignments and examinations.

Private Tutoring

2022–present **Private Mathematics Tutor**

Independent

- Four years of individual mathematics tutoring.
- Offer tutoring in calculus, linear algebra, and introductory real analysis.

Last updated: November 2, 2026